

One-dimensional Turbulence Model (notes)

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Problem 3

Write the equation of motion for a velocity fluctuation.

We start by recalling the equation of motion for momentum, after performing Reynold's decomposition and splitting components into their mean and fluctuating parts...

$$\partial_t(\bar{u}_j + u'_j) + (\bar{u}_i + u'_i)\partial_i(\bar{u}_j + u'_j) = -\frac{1}{\rho}\partial_j(\bar{P} + P') + \nu\partial_k\partial_k(\bar{u}_j + u'_j)$$

$$\partial_t\bar{u}_j + \partial_t u'_j + \bar{u}_i\partial_i\bar{u}_j + \bar{u}_i\partial_i u'_j + u'_i\partial_i\bar{u}_j + u'_i\partial_i u'_j = -\frac{1}{\rho}\partial_j\bar{P} - \frac{1}{\rho}\partial_j P' + \nu\partial_k\partial_k\bar{u}_j + \nu\partial_k\partial_k u'_j \quad (1)$$

Now we can subtract our RANS equation for momentum from this equation to isolate the turbulent components (u'). Recall our RANS equation for momentum reads

$$\partial_t\bar{u}_j + \bar{u}_i\partial_i\bar{u}_j = -\frac{1}{\rho}\partial_j\bar{P} + \nu\partial_k\partial_k\bar{u}_j - \partial_i\overline{u'_j u'_i}. \quad (2)$$

Subtracting Equation (2) from Equation (1) yields

$$\partial_t u'_j + \bar{u}_i\partial_i u'_j + u'_i\partial_i\bar{u}_j + u'_i\partial_i u'_j = -\frac{1}{\rho}\partial_j P' + \nu\partial_k\partial_k u'_j + \partial_i\overline{u'_j u'_i}. \quad (3)$$

$$\partial_t u'_j + \bar{u}_i\partial_i u'_j = -\frac{1}{\rho}\partial_j P' + \nu\partial_k\partial_k u'_j + \partial_i\overline{u'_j u'_i} - u'_i\partial_i\bar{u}_j - u'_i\partial_i u'_j. \quad (4)$$

Derive equation for Reynold's Stress Transport

We multiply this equation by u'_k , expanding this into a 3x3 tensor equation.

$$u'_k \partial_t u'_j + u'_k \bar{u}_i \partial_i u'_j = -u'_k \frac{1}{\rho} \partial_j P' + u'_k \nu \partial_z \partial_z u'_j + u'_k \partial_i \overline{u'_j u'_i} - u'_k u'_i \partial_i \bar{u}_j - u'_k u'_i \partial_i u'_j. \quad (5)$$

Reynold's Average:

$$\langle u'_k \partial_t u'_j \rangle + \langle u'_k \bar{u}_i \partial_i u'_j \rangle = -\langle u'_k \frac{1}{\rho} \partial_j P' \rangle + \langle u'_k \nu \partial_z \partial_z u'_j \rangle + \langle u'_k \partial_i \overline{u'_j u'_i} \rangle - \langle u'_k u'_i \partial_i \bar{u}_j \rangle - \langle u'_k u'_i \partial_i u'_j \rangle. \quad (6)$$

The divergence of the averaged Reynold's Stress goes to zero, as it's an averaged quantity multiplied by a fluctuating value

$$\langle u'_k \partial_t u'_j \rangle + \langle u'_k \bar{u}_i \partial_i u'_j \rangle = -\langle u'_k \frac{1}{\rho} \partial_j P' \rangle + \langle u'_k \nu \partial_z \partial_z u'_j \rangle + \langle u'_k u'_i \partial_i \bar{u}_j \rangle - \langle u'_k u'_i \partial_i u'_j \rangle. \quad (7)$$

We now take a second equation, flip the indices

$$\langle u'_j \partial_t u'_k \rangle + \langle u'_j \bar{u}_i \partial_i u'_k \rangle = -\langle u'_j \frac{1}{\rho} \partial_k P' \rangle + \langle u'_j \nu \partial_z \partial_z u'_k \rangle + \langle u'_j u'_i \partial_i \bar{u}_k \rangle - \langle u'_j u'_i \partial_i u'_k \rangle \quad (8)$$

and add these together. To reduce the messiness, we will first look at the LHS,

$$\langle u'_j \partial_t u'_k \rangle + \langle u'_j \bar{u}_i \partial_i u'_k \rangle + \langle u'_k \partial_t u'_j \rangle + \langle u'_k \bar{u}_i \partial_i u'_j \rangle. \quad (9)$$

To simply, we apply the reverse product rule (ie, given we know)

$$\frac{\partial}{\partial t} \langle u'_k u'_j \rangle = \langle u'_k \frac{\partial}{\partial t} u'_j \rangle + \langle u'_j \frac{\partial}{\partial t} u'_k \rangle$$

and

$$\bar{u}_i \frac{\partial}{\partial x_i} \langle u'_k u'_j \rangle = \bar{u}_i \langle u'_k \frac{\partial}{\partial x_i} u'_j \rangle + \bar{u}_i \langle u'_j \frac{\partial}{\partial x_i} u'_k \rangle,$$

we can rewrite this LHS as

$$\partial_t \langle u'_k u'_j \rangle + \bar{u}_i \partial_i \langle u'_k u'_j \rangle.$$

Now onto the fun bit... the RHS. Currently we have

$$-\langle u'_k \frac{1}{\rho} \partial_j P' \rangle - \langle u'_j \frac{1}{\rho} \partial_k P' \rangle + \langle u'_k \nu \partial_z \partial_z u'_j \rangle + \langle u'_j \nu \partial_z \partial_z u'_k \rangle + \langle u'_k u'_i \partial_i \bar{u}_j \rangle + \langle u'_j u'_i \partial_i \bar{u}_k \rangle - \langle u'_k u'_i \partial_i u'_j \rangle - \langle u'_j u'_i \partial_i u'_k \rangle.$$

Some of this follows the same pattern (reverse product rule), allowing us to expand our fluctuating pressure term. Consider that

$$\frac{\partial}{\partial x_j} \langle u'_k P' \rangle = \langle u'_k \partial_j P' \rangle + \langle P' \partial_j u'_k \rangle.$$

We substitute

$$\begin{aligned} -\langle u'_k \frac{1}{\rho} \partial_j P' \rangle &= \frac{1}{\rho} (\langle P' \partial_j u'_k \rangle - \partial_j \langle u'_k P' \rangle) \\ -\langle u'_j \frac{1}{\rho} \partial_k P' \rangle &= \frac{1}{\rho} (\langle P' \partial_k u'_j \rangle - \partial_k \langle u'_j P' \rangle) \end{aligned}$$

Next we identify the advection of mean flow by the Reynolds stress, i.e., the terms

$$\langle u'_k u'_i \partial_i \bar{u}_j \rangle + \langle u'_j u'_i \partial_i \bar{u}_k \rangle.$$

We won't rearrange these terms for now, since they're already in a form involving an averaged Reynold's stress. The viscous dissipation, however, looks like an interesting challenge,

$$\nu \langle u'_k \partial_z \partial_z u'_j \rangle + \nu \langle u'_j \partial_z \partial_z u'_k \rangle.$$

We would like to write this as the diffusion of the Reynolds stress. We expect such diffusion to look like

$$\nu \partial_z \partial_z \langle u'_k u'_j \rangle,$$

which would expand (ignoring the coefficient) as

$$\begin{aligned} \partial_z (\partial_z \langle u'_k u'_j \rangle) &= \partial_z (\langle u'_k \partial_z u'_j \rangle + \langle u'_j \partial_z u'_k \rangle) \\ &= \langle u'_k \partial_z \partial_z u'_j \rangle + \langle \partial_z u'_j \partial_z u'_k \rangle + \langle \partial_z u'_j \partial_z u'_k \rangle + \langle u'_j \partial_z \partial_z u'_k \rangle. \end{aligned}$$

We relate this to the term in the equation by substituting

$$\nu \langle u'_k \partial_z \partial_z u'_j \rangle + \nu \langle u'_j \partial_z \partial_z u'_k \rangle = \nu \partial_z \partial_z \langle u'_k u'_j \rangle - 2\nu \langle \partial_z u'_j \partial_z u'_k \rangle.$$

The last terms we want to deal with are the advection of turbulent fluctuations by the Reynolds stress itself,

$$-\langle u'_k u'_i \partial_i u'_j \rangle - \langle u'_j u'_i \partial_i u'_k \rangle.$$

We recognize this pattern, and can apply reverse-product-rule. The last step is due to incompressibility (allows us to put u_i inside the derivative).

$$-\langle u'_i u'_k \partial_i u'_j \rangle - \langle u'_i u'_j \partial_i u'_k \rangle = -\langle u'_i \partial_i u'_k u'_j \rangle = -\langle \partial_i u'_k u'_j u'_i \rangle$$

Now we can put together all the steps. For notation in this last step, I am using capital U to denote an averaged velocity (instead of an overbar).

$$\partial_t \langle u'_k u'_j \rangle + U_i \partial_i \langle u'_k u'_j \rangle = \quad (10)$$

% triple correlation

$$-\partial_i \langle u'_k u'_j u'_i \rangle + \langle u'_k u'_i \rangle \partial_i U_j + \langle u'_j u'_i \rangle \partial_i U_k$$

% viscous stress

$$+\nu \partial_z \partial_z \langle u'_k u'_j \rangle - 2\nu \langle \partial_z u'_j \partial_z u'_k \rangle + \frac{1}{\rho} (P' \langle \partial_j u'_k + \partial_k u'_j \rangle - \partial_j \langle u'_k P' \rangle - \partial_k \langle u'_j P' \rangle)$$

We can define all these terms and define them from a physics perspective. Starting left to right,

1. Storage / time variation of the Reynold's stress
2. Advection of the Reynold's stress by the mean flow
3. Triple Correlation: transport of Reynold's stress by turbulent motions.
4. Advection of flow by Reynold's stress. This is also seen as a production term, as it captures the conversion of turbulence from the mean flow through shear.
5. (shear production)
6. Diffusion of mometum from viscosity / molecular interactions.
7. This is a very exciting term! It is the viscous dissipation term and it *directly destroys* Reynold's stress through conversion to heat.

8. These next two terms are very interesting. Franko spent a long time talking to me about this one before he graduated, and said if I understood this term, I would “understand some grand part of turbulence.” Whether or not that is true has yet to be seen. However, this is the pressure-strain correlation. It resembles the rate-of-strain tensor, and similar to S_{ij} , this tensor is traceless. This term cannot create or destroy turbulence, however, it often performs what is called “pressure scrambling.” One of these terms will anisotropize the turbulence (“fast term”) and the other isotropizes the flow (“slow term”).
9. These terms are transport by a pressure-fluctuation correlation term.

The TKE Equation can be expressed as

$$\frac{\partial k}{\partial t} + \bar{U}_j \quad (11)$$

The density profile is calculated according to

$$\rho = \rho_0(1 - \alpha(C - T_0)) \quad (12)$$

The Mellor-Yamada closure

We start with the continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho U_i) = 0 \quad (13)$$

where flow is considered to be divergent-free, or solenoidal.

Meanwhile, the Reynolds-Averaged Navier Stokes reads:

$$\frac{\partial \bar{U}}{\partial t} + \bar{U}_j \frac{\partial \bar{U}_k}{\partial x_j} + \rho \epsilon_{jkl} f_k U_l = \frac{\partial}{\partial x_k} \overline{u_k u_j} - \frac{\partial \bar{P}}{\partial x_j} - g \hat{e}_3 \quad (14)$$

The Boussinesq approximation is invoked to represent that density fluctuations affect buoyancy but no other transport/state terms. These fluctuations are written $\beta\theta$, where β is the coefficient of thermal expansion.

Mellor and Herring (1973) uses the following equation:

$$\left\langle \frac{p}{\rho} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right\rangle = -\frac{q}{3l_1} (\overline{u_i u_j} - \frac{\delta_{ij}}{3} q^2) + C_1 q^2 \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) \quad (15)$$

We are establishing equations for the following: $\overline{u_k u_i u_j}$, $\overline{p u_j}$, $\overline{u_i u_j \theta}$, and $\overline{p \theta}$

Math Note The velocity gradient $\frac{\partial U_j}{\partial x_i}$ can be decomposed into symmetric and antisymmetric parts,

$$\frac{\partial U_j}{\partial x_i} = S_{ij} + \Omega_{ij}$$

where the symmetric rate-of-strain tensor is

$$S_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right)$$

and the asymmetric part is,

$$\Omega_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i} \right)$$

Parameterization

$$\langle u_k u_i u_j \rangle = \frac{3}{5} lq S_q \left(\frac{\partial \langle u_i u_j \rangle}{\partial x_k} + \frac{\partial \langle u_i u_k \rangle}{\partial x_j} + \frac{\partial \langle u_j u_k \rangle}{\partial x_i} \right) \quad (16)$$

In this equation S_q is a dimensionless number.

Sm :

$$\frac{A_1}{A_2} \frac{B_1(\gamma - C_1) - (B_1(\gamma_1 - C_1) + 6(A_1 + 3A_2))R_f}{B_1\gamma_1 - (B_1(\gamma_1 + \gamma_2) - 3A_1)R_f} \quad (17)$$

where

$$\gamma_1 = \frac{1}{3} - \frac{2A_1}{B_1}$$

and

$$\gamma_2 = \frac{B_2}{B_1} - \frac{6A_1}{B_1}$$

Sh:

$$S_h = \frac{A_2(1 - \frac{6A_1}{B_1})}{1 - (3A_2B_2G_h + 18A_1A_2G_h)} = \frac{A_2(1 - \frac{6A_1}{B_1})}{(1 - 3A_2G_h(B_2 + 6A_1))}$$

$$S_m = \frac{A_1(1 - 3C_1 - \frac{6A_1}{B_1}) + S_H G_h(18A_1 + 9A_1A_2)}{(1 - 9A_1A_2G_h)}$$

Richardson's number:

$$G_H = -\frac{L^2}{Q^2} \frac{g}{\rho_0} \frac{\partial}{\partial z}$$